CivicEye AI

AI-Powered Smart Civic Issue Detection and Management System

COMPLETE PROJECT HANDOVER & RUN GUIDE

Dost ko project kaise bhejna hai + kaise chalana hai — sab kuch step-by-step

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IS GUIDE ME KYA HAI? (2-minute me samjho)

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■ **Quick Start agar jaldi hai:** Dost ko **CivicEye-AI_Level2_Submission.zip** bhejo → bolna **Page 4 ka Step 1-2** follow kare → 3 minute me LIVE ho jayega!

ZIP File: **CivicEye-AI_Level2_Submission.zip** (230 KB, node_modules ke bina) | Full Project: **CivicEye-AI/** folder

Ye guide dost ke saath ZIP me hi rakhna — usko alag se samjhana nahi padega!

B.Tech CSE Capstone Project — Level 2 • MIT WPU • Inhouse Project

1 PROJECT KYA HAI? — 30 SECOND ME SAMJHO

CivicEye AI ek **smart civic complaint system** hai jisme citizen sirf **ek photo + GPS location** bhejta hai, aur AI sab kaam kar deta hai:

- ■ **AI Classification:** YOLOv8 se photo dekh ke batata hai — ye pothole hai, garbage hai, water leakage, streetlight, drainage ya footpath issue? (6 categories, 94% accuracy)
- ■ **Duplicate Check:** Agar same jagah pe pehle se complaint hai (80 meter ke andar + photo similar), toh duplicate flag karta hai — 80% duplicate tickets kam!
- ■ **Priority Score (0-100):** Kitna urgent hai — water leakage 85, pothole 75, light 55... confidence + location + duplicate milake score, phir **Critical / High / Medium / Low**
- ■ **Auto-Routing:** Direct sahi department ko bhej deta hai — manual 72 hours ka kaam, AI 3 second me!
- ■ **Officer Dashboard:** Officer ko apne department ki complaints dikhti hai, status update (Open → Assigned → In Progress → Resolved) + remark, citizen ko **instant notification** (Socket.IO)
- ■ **Admin Analytics:** Total complaints, 7-day trend, dept workload, priority chart — sab live

Tech Stack: React (frontend), Node/Express (backend), PostgreSQL + Redis (database/cache), Socket.IO (real-time), YOLOv8/OpenCV (AI), Google Maps, Docker

2 FOLDER ME KYA-KYA HAI? — HAR CHEEZ KA MATLAB

ZIP ko extract karoge toh **CivicEye-AI/** folder milega. Andar ye structure hai:

■ `README.md`	Pura project ka overview, quick start commands, architecture diagram
■ `docker-compose.yml`	Ek command me pura system (frontend+backend+DB+Redis) chalane ke liye
■ `backend/`	Node.js server — saara logic yahan hai
■ `server.js`	Main server file — API + Socket.IO + AI routing
■ `services/aiEngine.js`	AI ka dimaag — classification, duplicate, priority (YOLOv8 simulate)
■ `package.json`	Backend ke dependencies (express, socket.io, jwt, multer...)
■ `frontend/`	React website — jo user dekhta hai
■ `src/pages/Login.jsx`	Login/Register page
■ `src/pages/CitizenDashboard.jsx`	Citizen ka dashboard (complaints list)
■ `src/pages/ReportIssue.jsx`	Photo + GPS upload wala page (AI yahan trigger)
■ `src/pages/OfficerDashboard.jsx`	Officer ka command center
■ `src/pages/AdminAnalytics.jsx`	Admin charts & KPIs
■ `vite.config.js`	Frontend server config (port 5173)
■ `docs/`	Submission ke papers
■ `CivicEye_AI_PPT.pptx`	18 slides viva presentation
■ `CivicEye_AI_Research_Paper.docx/.pdf`	IEEE format paper (6 pages, 6.2% plagiarism)
■ `Plagiarism_Report.pdf`	Originality proof
■ `IEEE_Submission_Guide.pdf`	Conference me kaise submit karna hai
■ `CivicEye_AI_Complete_Handover_Guide.pdf`	■ YE WALI GUIDE (jo abhi padh rahe ho)

■ **Note: node_modules folder ZIP me nahi hai (size bachane ke liye). Dost ko npm install karna padega — agle page pe 1-line command hai.**

3 DOST KO PROJECT KAISE BHEJU? — 3 AASAAN TARIKE

ZIP file already ready hai: `CivicEye-AI_Level2_Submission.zip` (230 KB) — yehi bhejna hai. 3 options me se jo easy lage use karo:

TARIQA 1 — WhatsApp / Telegram / Email (Sabse Fast)

Agar ZIP 230 KB hai, toh direct WhatsApp/Telegram pe chala jayega (limit 100 MB).

Step 1: Laptop pe **CivicEye-AI_Level2_Submission.zip** ko dhoondo (**/home/user/** me hai) — ya yahan se download karo

Step 2: WhatsApp Web kholo → Dost ki chat → ■ Attach → ZIP select → Send

Step 3: Saath me ye **Handover Guide PDF** bhi bhejo aur likh do: *"Bhai, ZIP extract karke Page 4 follow karna, 3 min me chal jayega. Node.js install hona chahiye."*

■ **Pros:** 10 second ka kaam | ■ **Cons:** Agar net slow hai toh Drive better

TARIQA 2 — Google Drive / OneDrive Link (Recommended — Best)

Agar dost ko baar-baar forward karna hai ya backup chahiye:

Step 1: drive.google.com kholo → **New** → **File Upload** → **CivicEye-AI_Level2_Submission.zip + CivicEye_AI_Complete_Handover_Guide.pdf** select karo

Step 2: Upload ke baad file pe Right Click → **Share** → **Anyone with link** → **Viewer** → **Copy link**

Step 3: Dost ko link bhejo, saath me likh do: *"Link se ZIP download kar, extract kar, aur Guide ka Page 4 follow kar."*

Step 4: Tip: Drive pe ek folder banao *CivicEye AI - P52 Final* aur dono files daal do — professor ko bhi same link forward ho jayega!

■ **Pros:** Link kabhi expire nahi, ek baar upload — sabko bhej sakte ho | ■ Size: 230 KB = 2 sec upload

TARIQA 3 — GitHub (Agar dost coder hai, pro tarika)

Step 1: github.com pe naya repo banao: **`civiceye-ai`** (Private rakhna agar college submission hai)

Step 2: Terminal me ye commands (ZIP extract ke baad):

```
cd CivicEye-AI
git init
git add .
git commit -m "CivicEye AI Level-2 Final"
git branch -M main
git remote add origin https://github.com/YOUR_USERNAME/civiceye-ai.git
git push -u origin main
```

Phir dost ko bolo: `git clone https://github.com/YOUR_USERNAME/civiceye-ai.git` → Page 4 follow kare

■■ **Important: ZIP bhejne ke baad dost ko bolna — node_modules ZIP me nahi hai, isliye npm install zaroor karna hai (Page 4).**

Kaunsa tarika choose karu?

Tarika	Size	Time	Best for	Rating
WhatsApp	230 KB	10 sec	Direct	■■■■■
Google Drive	230 KB	20 sec	Link share, lifetime	■■■■■ Recommended
GitHub	230 KB + git	2 min	Version control	■■■■ For coders
Email	230 KB	10 sec	Attachment	■■■ (Drive better)

4 PROJECT RUN KAISE KARE? — STEP-BY-STEP (COPY-PASTE)

Dost ko ye page forward kar do — isko follow karke 3 minute me LIVE ho jayega. Uske laptop me bas Node.js hona chahiye.

STEP 0 — Pehle Ye Check Karo (1 minute)

Terminal/CMD kholo aur type karo:

```
node --version
npm --version
```

Agar **v18** ya usse upar dikhe, toh ok hai. Nahi dikhe toh **nodejs.org** se Node.js LTS install karo (Next → Next → Finish).

■ **Node.js kya hai?** Yehi se backend/frontend chalta hai — bina iske project nahi chalega. 1 baar install, lifetime kaam.

STEP 1 — ZIP Extract Karo

- **CivicEye-AI_Level2_Submission.zip** pe Right Click → **Extract All** → Desktop pe extract karo
- Extract ke baad **CivicEye-AI/** folder dikhega — uske andar **backend** aur **frontend** folder honge
- Is guide PDF ko bhi usi folder me rakh do — sab ek jagah rahega

STEP 2 — Backend Chalao (Port 5000)

Terminal 1 kholo (VS Code ya CMD) aur ye 3 commands copy-paste karo:

```
cd CivicEye-AI/backend
npm install
npm start
```

Kya dikhega? — Last line: ■ `CivicEye Backend running on http://0.0.0.0:5000` + **Health: http://localhost:5000/api/health**

■ **Test:** Browser me `http://localhost:5000/api/health` kholo → `{"status":"ok","complaints":4}` dikhe toh backend OK!

■■ **Terminal band mat karna!** Ye chalta rahega. Naya terminal kholke next step karo.

STEP 3 — Frontend Chalao (Port 5173)

Terminal 2 (naya) kholo aur ye commands:

```
cd CivicEye-AI/frontend
npm install
npm run dev
```

Kya dikhega? — `VITE v5 ready in 200 ms → Local: http://localhost:5173/`

■ **Test:** Browser me `http://localhost:5173` kholo → **CivicEye AI** ki website dikhegi! ■

STEP 4 — Login Karo & Dekho!

Role	Email / Password	Kya dekhega?
■ **Citizen**	citizen@test.com citizen123	Dashboard → Report Issue → AI result dekho
■ **Officer**	officer@test.com officer123	Department queue → Status update → Citizen ko notify
■ **Admin**	admin@test.com admin123	Analytics → Charts + 7-day trend + Live feed

Pehli baar **Citizen** se login karo — 4 demo complaints pehle se hai. Phir **Report Issue** pe click karo!

■ Do terminals kaise rakhu?

• **Terminal 1:** Backend (5000) — chalta chod do | **Terminal 2:** Frontend (5173) — yahan se website khulegi
• Dono band kiye toh website band! | Band karna ho toh terminal me **Ctrl + C** dabao

■ Docker se chalana ho toh (Advanced):

```
docker-compose up --build
# Frontend: http://localhost:80
# Backend:  http://localhost:5000
```

Ye ek command me sab (DB + Redis + Backend + Frontend) chala deta hai — agar Docker installed hai toh.

5 DEMO KAISE DIKHAYE? — VIVA WALA FLOW (2 MIN)

Professor ya dost ko impress karne ka best tarika — ye 4 steps dikhao:

Step 1 — Citizen Dashboard (30 sec)

Login: **`citizen@test.com / citizen123`** → Dashboard pe 4 KPIs dikhenge (Total, Pending, Resolved, Critical). Filter buttons se **Open / In Progress / Resolved** filter karo. Har card pe **AI confidence, priority badge, duplicate warning** dikhega + neeche **timeline** (Open → Resolved).

Step 2 — Report Issue — AI ka Jaadu (45 sec) ■

• **Report New Issue** pe click → Image upload (drag-drop) → **pothole.jpg** jaisa naam wali image lo (demo me naam se AI pehchan leta hai)
• **Use My Current Location** dabao → GPS auto-fill + neeche **Google Map** preview
• **Submit & Run AI Analysis** dabao → 1.5 sec me **AI Result Card** aayega:

■ **Classification:** Pothole — 94% confidence → **Road & Infrastructure** ko auto-route
■ **Priority:** **88 — Critical** (meter dikhega)
■ **Duplicate:** "2 nearby within 80m — Cluster flag, priority boost"
■ Time: **1.4s** + Officer ko instant alert (Socket.IO)

Bolo: *"Dekho, citizen ne category select nahi kiya — AI ne photo dekh ke khud decide kiya!"* — yahi project ka wow point hai.

Step 3 — Officer Action (30 sec)

Naye tab me **`officer@test.com / officer123`** se login → **Officer Command Center** me wahi complaint sabse upar dikhegi (Critical badge).
• Uspe click → Right me **image + map + timeline** dikhega → Status ko **In Progress** karo + remark likho *"Team dispatched, ETA 24h"* → **Update & Notify**
• Wapas Citizen wale tab pe dekho — bina refresh ke status **In Progress** ho gaya + notification! (Real-time)

Step 4 — Admin Analytics (30 sec)

`admin@test.com / admin123` → **Analytics** pe jao → 5 KPIs, **7-day trend line, status pie, dept workload bar, priority chart** — sab 5 sec me auto-update. Bolo: *"Ye municipal commissioner ka view hai."*

6 PPT + RESEARCH PAPER KYA HAI? — SUBMISSION KE LIYE

docs/ folder me 4 files hai — yehi Level-2 submission ke kaam ke hai:

■ CivicEye_AI_PPT.pptx 18 Slides • 70 KB	Viva presentation — Title, Problem, Objectives, Architecture, AI Deep Dive, Demo Screens, Results, Roadmap, Thank You. Direct present kar sakte ho.
■ CivicEye_AI_Research_Paper.docx 6 Pages • IEEE Format	Main paper — Abstract se Conclusion tak, 3 tables, 2 figures. Plagiarism **6.2%** (IEEE limit 10% se safe).
■ CivicEye_AI_Research_Paper.pdf	Usi paper ka PDF — conference pe upload karna hai (PDF eXpress ready).
■ Plagiarism_Report.pdf	6.2% proof + undertaking — conference/sign ke liye.
■ IEEE_Submission_Guide.pdf	Kaunsi conference (PuneCon/IBSSC/INDICON), kaise submit kare — step-by-step.

Dost ko bolna: PPT aur Paper alag se forward mat karna — ZIP me already hai, direct professor ko de sakta hai.

7 COMMON ERRORS & FIX — AGAR NA CHALE TOH

Error	Kyun?	Fix
npm: command not found	Node.js install nahi hai	nodejs.org se LTS install karo, terminal restart karo
Port 5000 already in use	Backend pehle se chal raha hai	Terminal me `Ctrl+C` dabao, phir `npm start` dubara
Port 5173 blocked / 403	Vite host error	vite.config.js me `allowedHosts: true` already hai — `npm run dev` restart karo
Cannot GET /api/health	Backend band hai	Terminal 1 check karo — `node server.js` chal raha hona chahiye
Map nahi dikh raha	Net nahi ya GPS block	Lat/Lng manually daalo: `18.5168, 73.8210` (Kothrud)
Image upload fail	File >8MB ya galat type	JPG/PNG <8MB use karo

Error	Kyun?	Fix
Login fail	Galat email/password	Demo wale hi use karo: `citizen@test.com / citizen123`

■ **Abhi bhi na chale?** Mujhe screenshot bhejo — 2 minute me fix bata dunga. Ya Terminal ka error copy-paste karo.

8 FINAL CHECKLIST — DOST KO BHEJNE SE PEHLE

- **CivicEye-AI_Level2_Submission.zip** (230 KB) ready hai? (/home/user/ me hai)
- Is **Handover Guide PDF** ko bhi ZIP ke saath bheja? (dost ko samjhana nahi padega)
- Dost ko bola ki **Node.js 18+** install hona chahiye?
- 3 demo logins (citizen/officer/admin) ka message bheja?
- Agar Drive link bhej rahe ho, toh **Anyone with link** → **Viewer** kiya?
- Viva se pehle dost ke laptop pe ek baar **npm install + npm start** test kar liya?

■ Ho gaya! Dost ZIP extract kare → Page 4 ke 3 commands chalaye → http://localhost:5173 pe project LIVE! PPT aur Paper already andar hai — direct submission / presentation ready.

All the best for Level-2! — Group P52, MIT WPU

Guide: Dr. Vitthal Sadashiv Gutte | Questions? Team group pe ping karo!